Unit 2: Application of Core Principles of Chemistry

Topics 2.3, 2.4 and 2.5 – Shapes of molecules and ions, intermediate bonding and bond polarity, intermolecular forces	
Electron-pair repulsion theory shows how chemists can make generalisations and use them to make predictions.	
2.3c	Students use the electron-pair repulsion theory to predict the shapes of unfamiliar molecules. **HSW 1**
2.3e	The work on nanochemistry could include colloid chemistry and the increased reactivity of nanoparticles compared to larger ones. Also in the cosmetics industry the use of nano-sized particles in creams etc so that they can be more easily absorbed through the skin, as in moisturisers, and particles that are smaller than the wavelength of light and are therefore too small to see, and can be used in sunscreen creams but are transparent to light rather than opaque. This could raise issues about risks to and implications for humans. **HSW 9, HSW 10**
2.4a and 2.4d	This topic raises the issue of the bonding model as applied to actual compounds and can lead students to consider the limitations of the model and the need to consider other ideas to explain observed phenomena. **HSW 1, HSW 7**
2.5b and 2.5d	Use of the bonding models to explain observed phenomena. **HSW 1**

Topics 2.6 and 2.7 – Redox, the periodic table – groups 2 and 7	
Chemists rationalise a great deal of information about chemical changes by using theory to categorise reagents and types of chemical change. This is illustrated by the use of inorganic and organic examples in this unit.	
2.6a	Students use the concept of oxidation number to classify reactions and to help their understanding of changes at the atomic level. **HSW 1**
2.7	This topic on the periodic table can be used to illustrate the ways in which chemists collate data, search for patterns, reduce the need to remember every single reaction of every compound and predict the chemistry of unfamiliar compounds such as those of astatine. **HSW 1, HSW 5**
2.7.1h and 2.7.2c	Using the technique of volumetric analysis provides an opportunity for students to evaluate their own methodology to include the limitations of the apparatus, accuracy of measurement, types of errors and uncertainty of the final result. The difference between reliability and uncertainty could be explored, including the idea that repeating an experiment with the same apparatus will not improve the accuracy, only the reliability and the need to appreciate that an uncertainty in a final result can have implications for how the information may be used. **HSW 5 with potential for HSW 12**
Topics 2.8 and 2.9 – Kinetics, chemical equilibria	
The use of models in chemistry is illustrated by the way in which the Maxwell-Boltzmann distribution and collision theory can account for the effects of temperature on the rates of chemical reactions.	
2.8b	The collision theory provides a model that students can apply to help their understanding of how reactions happen at the molecular level and explain the effect of changing conditions on the rate of a reaction. **HSW 1**
2.8c	The Maxwell-Boltzmann curves provides students with a mathematical model that allows them to understand and predict the effect of changing conditions on the rate of a reaction. **HSW 1**
2.8d and 2.8e	Use of the concept of activation energy to explain changes in rate of reactions. **HSW 1**
2.9c	The equilibrium model and the effect of changes on the position of equilibrium can be applied to industrial situations. This can lead into discussions of atom economy and the ways in which industrial chemists and chemical engineers manipulate the conditions in a reaction to maximize yield and minimise waste of raw materials. Examples might include recycling of unreacted gases and finding new catalysts that work at lower temperatures. **HSW 9**

Topics 2.10, 2.11 and 2.12 – Organic chemistry, mechanisms, mass spectra and IR	
The unit shows how chemists study chemical changes on an atomic scale and propose mechanisms to account for their observations.	
2.10.2f	Discussion of the use of fire retardants and a consideration of the benefits against the risks involved with the use of halogenoalkanes. **HSW 9**
2.11a to 2.11e	The work on mechanisms provides an example of the advantages of classifying reactions and models and will help students' understanding of what is happening at the molecular level. **HSW 1**
2.11f	Students can apply the model to deepen their understanding of how reactions proceed at the molecular level including substitution in halogenoalkanes. **HSW 1**
2.11g	Consideration of the depletion of the ozone layer, NO emissions from cars and the problem of exhaust NO from high flying aircraft could be used as a starting point for discussions on how understanding the mechanism of a reaction can help scientists to propose solutions to environmental problems such as the role catalytic converters in cars can play in the reducing of NO emissions. Students should understand the difference between greenhouse gases and those associated with the ozone layer depletion but also that some are the same. **HSW 2, HSW 3, HSW 5, HSW 8, HSW 10, HSW 12**
2.12d and 2.13b	The model of polarity allows chemists to understand why some gases present problems in terms of global warming while others do not. **HSW 2**
Topics 2.13 – Green chemistry	
2.13	This topic provides an opportunity to bring together areas where chemists are involved in attempting to solve environmental problems by the application of theories and new technologies and are helping the public to understand the implications of the decisions that they make. This includes efforts by industry such as: ■ consideration of renewable resources ■ making industrial processes more environmentally friendly ■ using more efficient energy sources for reactions ■ reducing waste. Students should consider the concept of carbon neutrality and explore the need to consider all stages of a process. Students should identify examples of the way scientific information has helped society to change. **HSW 9, HSW 10, HSW 11, HSW 12**